

# A Cosmological-Constant Acceleration Scale and its Modified-Inertia Completion

**The MOND scale as a de Sitter–Unruh inertial effect:  $a_0 = c^2 \sqrt{\Lambda/32\pi}$ , its written covariant action, dark-matter-free lensing, and a falsifiable  $a_0(z)$**

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## Abstract

The MOND acceleration scale famously coincides with the cosmic acceleration scales,  $a_0 \sim cH_0 \sim c\sqrt{\Lambda}$  — a coincidence that  $\Lambda$ CDM treats as accidental. This paper takes it literally and develops it as a modified-**inertia** effect, not a modification of gravity: the acceleration at which a body’s inertial response to the de Sitter (dark-energy) vacuum departs from Newtonian is set by the vacuum energy density, giving

$$a_0 = \frac{c}{2} \sqrt{G\rho_\Lambda} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = \frac{cH_\Lambda}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} \approx 5.79, \quad a_0 \approx 9.4 \times 10^{-11} \text{ m s}^{-2}.$$

Two claims are made and separated cleanly. **(i) An empirical reframing.** At this  $a_0$  and a single stellar mass-to-light ratio, the SPARC radial-acceleration and baryonic Tully–Fisher relations are reproduced to  $\sim 0.11$  dex, with the interpolation  $\nu(y) = \sqrt{1 + 1/y}$  over-determined by the Deser–Levin de Sitter–Unruh temperature rather than fitted. **(ii) A covariant completion.** The explicit nonlocal modified-inertia action is written and carried to a definite boundary, reasoning throughout from the framework’s own premises — a **passive** reference frame, with the MOND content in the **matter** sector. On those premises the theory has a machine-verified closed constraint algebra (zero propagating frame degrees of freedom), a rigorously defined causal nonlocal operator, a causal ghost-free two-point sector, and it **evades the Cassini quadrupole** by  $\sim 7$  orders of magnitude where modified-gravity realizations of MOND fail at  $+6$  to  $+14\sigma$ ; because the frame is passive it carries no propagating tensor mode, so it satisfies the GW170817 graviton-speed bound trivially. **Gravitational lensing** is resolved without dark matter by a disformal photon metric built from the same passive frame: universal (one rule for all galaxies), Cassini-safe, ghost-free and causal. The one- and two-loop de Sitter **radiative structure is computed at the divergence level**:  $a_0$  is not renormalized, and no transverse aether kinetic term is generated in either the matter or the graviton sector, so the passive frame survives two loops at that level. **The one distinctive, falsifiable prediction** is that  $a_0$  evolves with the dark-energy density,  $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$ , declining to  $\approx 0.74 a_0(0)$  by  $z = 3$ ; the framework’s distinctive content dissolves back to ordinary MOND if dark energy proves to be a true cosmological constant. The MOND **sign**, the value of  $a_0$ , and the normalization  $Z$  are **posited inputs, not derived**; the Standard Model is untouched. This is a one-scale effective theory carried to a sharp, named boundary — explicitly **not** a theory of everything.

## 1. Introduction

Rotation curves of galaxies flatten below a characteristic acceleration  $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$  [Milgrom 1983; Lelli, McGaugh & Schombert 2016]. That scale sits within an order-one factor of both  $cH_0$  and  $c\sqrt{\Lambda}$ . In  $\Lambda$ CDM this is a numerical accident; in the modified-dynamics program it is a clue. Two

readings are possible. **Modified gravity** promotes a new field that sources the metric (AeST [Skordis & Złotnik 2021], the khronon [Blanchet & Skordis 2025], TeVeS); these bend light correctly but a dynamical vector/tensor sector generically fails solar-system tests. **Modified inertia** [Milgrom 1994, 1999] instead changes how bodies respond to a given gravitational field; it evades the solar-system problem but has historically lacked both a covariant action and a lensing account.

This paper develops the modified-inertia reading with the acceleration scale fixed by the **cosmological constant**. Section 2 states the scale; Section 3 gives the empirical anchor; Section 4 writes the covariant action and its verified structure; Section 5 resolves lensing; Section 6 reports the radiative structure; Section 7 gives the falsifiable prediction; Section 8 states the honest boundary. Every load-bearing claim is backed by a committed, runnable verification script; both admissible normalizations of  $a_0$  (“footings”) are carried throughout.

**Retraction honored.** Earlier versions of this program made theory-of-everything and Standard-Model claims that were publicly retracted (2026-06-23). They are not reasserted here. The claims of this paper are the  $a_0$  reframing and its covariant completion — nothing more.

## 2. The acceleration scale

Read  $a_0$  as the surface-gravity scale of the dark-energy horizon. With  $\rho_\Lambda = \Lambda c^2/8\pi G$  the dark-energy density and  $H_\Lambda = c\sqrt{\Lambda/3}$  its de Sitter rate,

$$a_0 = \frac{c}{2}\sqrt{G\rho_\Lambda} = c^2\sqrt{\frac{\Lambda}{32\pi}} = \frac{cH_\Lambda}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} \approx 5.79.$$

The scale is one order-one factor  $Z$  below the bare horizon acceleration  $cH_\Lambda$ . Two footings are admissible and are reported side by side everywhere: the **canonical** dark-energy footing  $\rho_{\text{DE}}/cH_\Lambda \Rightarrow a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$ , and the **total-density** footing  $\rho_{\text{tot}}/cH_0 \Rightarrow a_0 = 1.13 \times 10^{-10}$ . The interpolation between the Newtonian and deep-MOND regimes,

$$g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} a_0}, \quad \nu(y) = \sqrt{1 + 1/y}, \quad y = g_{\text{bar}}/a_0,$$

is the de Sitter–Unruh modified-inertia law of Milgrom [1999], fixed by the Deser–Levin temperature  $T(a) = (\hbar/2\pi k_B c) \sqrt{a^2 + (cH_\Lambda)^2}$  [Deser & Levin 1997] — the vacuum’s temperature acquires a floor from the horizon, and the departure from Newtonian inertia sets in when a body’s own acceleration drops to that floor. The  $O(1)$  coefficient  $32\pi$  (equivalently  $Z$ ) is a **posit**, not a theorem; it cancels in the falsifiable ratio  $a_0(z)/a_0(0)$  of Section 7.

## 3. The empirical anchor

On the SPARC sample [Lelli et al. 2016], at  $a_0 = 9.36 \times 10^{-11}$  and a single mass-to-light ratio  $\Upsilon_\star \approx 0.70$ , the radial-acceleration relation is reproduced with 0.108 dex scatter — marginally tighter than the standard MOND fit on the framework’s own  $\nu$  — and the baryonic Tully–Fisher relation  $v^4 = GM_{\text{bar}} a_0$  and the deep-MOND limit  $a = \sqrt{a_0 g_{\text{bar}}}$  are exact consequences. The RAR at this scale is convention-compatible but *non-diagnostic* of the specific value  $9.36 \times 10^{-11}$  versus  $1.13 \times 10^{-10}$  (the difference is absorbed by the interpolation and  $\Upsilon_\star$ ); the diagnostic content is the redshift evolution of Section 7, not the  $z = 0$  zero-point.

#### 4. The covariant action and its verified structure

With signature  $(-+++)$  the action is

$$S[x, g, u] = S_{\text{EH}}[g] + S_u[g, u] + S_{\text{matter}}[x, g, u],$$

with host general relativity unmodified, a unit-timelike frame constraint  $S_u = -\frac{1}{2}\sqrt{-g}\lambda(u^\mu u_\mu + 1)$  carrying **no aether kinetic term**, and the modified-inertia content in the matter kinetic sector,

$$S_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} \rho_m [s u^\mu K(\Box_u/a_0^2) u_\mu], \quad K(z) = \frac{\sqrt{1+4z}-1}{2\sqrt{z}}, \quad \Box_u f = u^a \nabla_a (u^b \nabla_b f).$$

Here  $u^\mu$  is a **passive**, horizon-anchored reference frame — an algebraic constitutive law, a Lorentz-violating gravitational-sector background, the cosmic rest frame — **not** a dynamical aether.  $K$  is non-entire with a branch cut and a single healthy pole;  $s = -1$  is the **MOND-sign posit** (see §8). Reasoning from these premises rather than the standard aether/MOND lens, the following are established, each machine-verified on both footings:

- **Worldline sector.** The circular-orbit reduction reproduces  $\nu(y) = \sqrt{1+1/y}$  to  $3 \times 10^{-13}$ ; Newton, deep-MOND, BTFR, the de Sitter-forced  $\sqrt{2}$ -weighted external-field kernel, and ghost-freedom (single healthy pole vs. the Ostrogradsky ghost of the local truncation) all follow.
- **Constraint structure.** The full curved-spacetime Dirac analysis closes at the secondary level; the unit-norm sector is genuinely second-class (its  $2 \times 2$  Dirac block has determinant  $4(u \cdot u)^2 \rightarrow 4$  on-shell), leaving **zero propagating frame degrees of freedom** — total propagating content is the two GR graviton polarizations. The concern that  $u$  sits *inside*  $\Box_u$  resolves negatively: the exact quadratic symbol at every order is  $S_n = (-1)^n k_\perp^2 k_0^{2n}$ , whose only root is  $k_0 = 0$  independent of spatial momentum — a transport-along- $u$  structure with no wave-cone and zero group velocity. The operator ordering of  $\Box_u$  is immaterial to this conclusion: the directional and covariantly-symmetrized orderings differ by a term  $\propto a^b \partial_b (u \cdot u)$  that vanishes by the unit-timelike constraint.
- **Operator.**  $K(\Box_u)$  is rigorously defined by the Borel functional calculus of the (essentially self-adjoint) directional operator, with a canonical Herglotz–Nevanlinna spectral representation carrying a unique positive Borel measure;  $\|K\| \leq 1$ , causal-retarded, defined on all of  $L^2$ . Its branch cut is the physical de Sitter–Unruh emission continuum, not a ghost.
- **Two-point sector.** The mixed matter–frame propagator is causal and ghost-free in closed form (retarded construction; Källén–Lehmann spectral positivity across the whole cut; principal symbol equal to the GR light-cone), and nonlinear classical stability holds.
- **Cassini and GW170817.** The modified-inertia realization evades the solar-system quadrupole: the anisotropic galactic external field enters only Saturn’s inertial response, deep-Newton-suppressed by  $\nu - 1 \approx 7 \times 10^{-7}$ , giving an  $\ell = 2$  quadrupole  $\sim 7$  orders below the Cassini ceiling, where the modified-gravity realization fails at  $+6$  to  $+14\sigma$ . Because the frame is passive it introduces no propagating tensor mode, so the graviton travels at  $c$  and the GW170817 bound is satisfied automatically — the constraint that eliminates a broad class of modified-gravity theories does not apply.

The Einstein-aether strong-coupling “wall” often invoked against covariant MOND is **mis-applied** here: it is a property of a *propagating* aether mode whose kinetic norm vanishes, and the framework’s frame has no such mode.

## 5. Gravitational lensing without dark matter

In modified inertia the dynamics come from the inertial response, not the metric, so a single-metric theory makes light — which has no inertia — track the *baryonic* metric and under-lens by the MOND factor  $\nu$ ; enhancing the one metric to compensate **double-counts** the dynamics (it would over-predict rotation curves), forcing pure modified gravity and reinstating the Cassini problem. The resolution is a **disformal photon metric** built from the passive frame,

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + B u_\mu u_\nu,$$

on which **light** propagates while matter keeps modified inertia in  $g$ . Only the disformal  $u_\mu u_\nu$  term bends light (a conformal factor is null-inert); fixing  $B$  by the same  $\nu$  the RAR uses makes lensing **universal** — one rule for every galaxy, not a per-galaxy fit — and **double-count-free by construction**, since the matter dynamics are untouched. It is **Cassini-safe**:  $B \sim (\nu - 1) \rightarrow a_0/2a \rightarrow 0$  at high acceleration, giving a PPN shift  $|\Delta\gamma| \sim 10^{-13}$  at the Sun, far below the  $2.3 \times 10^{-5}$  bound; and it is **ghost-free and causal** —  $\tilde{g}$  is Lorentzian ( $B \sim 3 \times 10^{-7} \ll 1$ ), photons are subluminal with respect to  $g$ , and the coupling is first-order in every dynamical field, hence Ostrogradsky-free. The disformal strength  $B$  is fixed non-locally by the framework’s **own** operator  $K(\square_u)$  (an AQUAL/Poisson lensing potential, momentum-conserving for any geometry; a local  $B(a/a_0)$  is its exact spherical limit) — so it is not a new ingredient. This recovers the “lensing tracks dynamics” property [Brouwer et al. 2021; Mistele & McGaugh 2024] from modified inertia, without the dynamical aether that fails Cassini.

## 6. Radiative structure on de Sitter

Using the framework’s own Herglotz measure, which converts the nonlocal vertex into a positive superposition of local massive resolvents, the one-loop divergence structure on de Sitter is computed exactly (Seeley–DeWitt, under a stated  $\rho_m = m^2\phi^2$  proxy for the quantum matter density):

- **$a_0$  is not renormalized**, additively or multiplicatively — there is no  $z^0$  tadpole at the exact-measure level, enforced by a sum rule  $\int d\mu(t)/|t| = K(\infty) - K(0) = 1$  (unit resolvent weight). The additive channel is protected to **all orders** by the exact shift symmetry  $T \rightarrow T + \text{const}$  (the frame depends only on  $\partial T$ ) together with the unit-timelike constraint, which collapses any surviving frequency-independent frame functional to a pure cosmological constant.
- **No transverse aether kinetic term is generated.** The dangerous graviton-frame mixing channel, real on de Sitter ( $R_{\mu\alpha\nu\beta}u^\alpha u^\beta = -H^2 P_\perp$  exactly), closes because the curvature commutators are algebraic ( $k_0^2 \rightarrow H^2$ , never  $k_\perp^2$ ). Extending to two loops, the transverse channel is closed at the divergence level in **both** sectors: in the **matter** sector by the geodesy identity (the linear frame vertex vanishes at every order) together with  $K(0) = 0$ ; and in the **graviton** sector by a complete census of two-graviton-loop topologies, in which the graviton’s transverse momentum is *rationed* (the frame’s external-momentum power is zero for all  $n$ , while a cone requires two), the transverse-traceless graviton-frame vertex vanishes, and the constrained shift sector stays instantaneous. So the passive frame (zero degrees of freedom) survives two loops at the divergence level.
- **Finite parts (matter sector).** The one-loop effective potential is bounded below (the exact bound  $\|K\| \leq 1$  with  $s = -1$  confines the effective mass-squared, precluding runaway), and the de Sitter infrared is regulated by the friction gap:  $\square_u$  is gapped at  $-9H^2/4$ , giving every loop field an effective mass  $\geq 3H/2$  that removes the usual secular-growth pathology. Dressed Källén–Lehmann positivity and KMS detailed balance survive the real one-loop convolution.

**Open at this level.** This is a *divergence-level* radiative account, not a finished quantum field theory: the one- and two-loop **finite** parts, genuinely higher loops, the disformal-metric variant of the matter coupling, and constraint-survival under loops all remain to be computed.

## 7. The falsifiable prediction

Because  $a_0$  is set by the dark-energy density, it is not constant:  $a_0(z) = c^2 \sqrt{\Lambda_{\text{eff}}(z)/32\pi} \propto \sqrt{\rho_{\text{DE}}(z)}$ . For a cosmological constant this is flat; for evolving dark energy it declines. With the DESI DR2 CPL best fit ( $w_0, w_a$ ) the density rises to a gentle peak  $\sim 4.5$  Gyr ago and then falls, so the framework predicts

$$a_0(z=3) \approx 0.74 a_0(0), \quad \text{with a } 6\% \text{ bump near } z \approx 0.4,$$

a **declining** acceleration scale. This is the theory's one distinctive, pre-registered signature, and it is falsifiable in both directions:

- **Confirms** if deep-MOND kinematics at  $z \approx 2-3$  (ELT/HARMONI, JWST/NIRSpec, ALMA) show  $a_0(z=3) \approx 0.74 a_0(0)$  tracking  $\sqrt{\rho_{\text{DE}}}$ ,  $> 3\sigma$  from both constant and rising alternatives; equivalently, high- $z$  discs sitting  $\approx -0.03$  dex below the  $z=0$  baryonic Tully–Fisher zero-point.
- **Falsifies** if  $a_0(z)$  is flat or rising, or — the standing hostage — if DESI DR3 drives  $w \rightarrow -1$ , in which case the distinctive content dissolves and the framework degenerates to ordinary MOND with no evolving scale.

The decisive near-term data are **DESI DR3** (the dark-energy equation of state), **Gaia DR4** (wide binaries and a Lorentz-violation dipole from the induced SME background), and **ELT-class high- $z$  rotation kinematics**.

## 8. The honest boundary

What is **derived / verified**: the interpolation shape (de Sitter–Unruh, over-determined); the worldline sector; the closed constraint algebra and zero frame degrees of freedom; the operator's rigorous definition; two-point causality and ghost-freedom; Cassini evasion and GW170817-safety; the disformal lensing construction (universal, Cassini-safe, ghost-free); and the divergence-level radiative structure in both sectors, on both footings.

What is **posited, not derived**: the **MOND sign**  $s = -1$  (a passive de Sitter vacuum supplies the anti-MOND sign; the active response that would fix it is not sourced without a pump — so the sign is a Machian input); the **value of  $a_0$**  and the normalization  $Z$  (a one-parameter effective theory, with  $Z$  not forceable); and there is **no Standard-Model bridge** — the particle sector is untouched.

What is **open**: the finite-part and higher-loop radiative structure; a first-principles condensate/AQUAL closure of the non-spherical lensing potential; and, above all, the empirical verdict on  $a_0(z)$ .

This is an effective, one-scale, falsifiable modified-inertia framework carried to a sharp and named boundary. It answers the objections that eliminate modified gravity (Cassini, GW170817) and the objection that traditionally eliminates modified inertia (lensing), with explicit, reproducible computations; and it stakes its distinctiveness on a single near-term measurement. It is **not** a finished quantum field theory and **not** a theory of everything, and it says so plainly.

## Reproducibility

All load-bearing claims are backed by committed, runnable scripts (repository `real_research/reviews/`), including the worldline sector and Cassini evasion, the constraint/operator/two-point analysis (`mi_formal_completion_2026/`), the disformal lensing construction and its ghost-freedom/locality checks, the one- and two-loop de Sitter radiative computations (`oneloop_lane*.py`, `twoloop_*`), the operator-ordering invariance check, and the full two-graviton-loop transverse-aether census (`graviton_census_2loop/`). Iterative companion: written MI action, Zenodo concept DOI 10.5281/zenodo.21253644.

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*Both  $a_0$  footings throughout. Each claim labeled derived / posited / open. The MOND sign,  $a_0$ , and  $Z$  are inputs; the Standard Model is untouched. No completeness or theory-of-everything claim is made.*